AI & COMPUTER-VISION BASED SURFACE & PRODUCT DEFECT INSPECTION

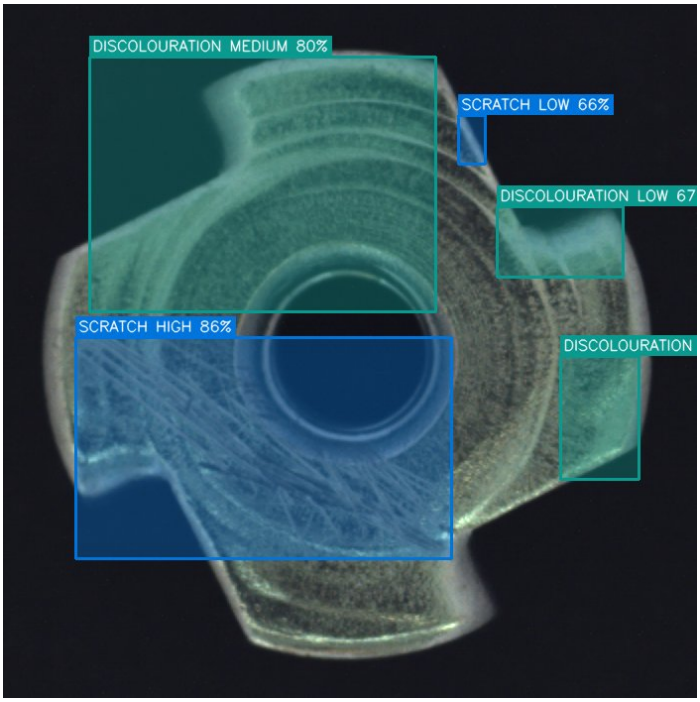
Automated Quality Inspection Report

Inspection ID:	DEF-2026-0332	**Date & Time:**	2026-11-01 20:43:58
Product / Part:	metal_nut	**Batch / Lot:**	-
Product Line:	GENERAL MANUFACTURED	**Station:**	-
Inspector:	Line Supervisor	**Outcome:**	**DEFECTIVE**

DEFECT SUMMARY

Defect Type	Count	Critical	High	Medium	Low
Discolouration	3	0	0	1	2
Scratch	2	0	1	0	1

ANNOTATED VISUAL EVIDENCE



DETECTION LOG

#	Type	Severity	Confidence	Location (px)	Detector	Review
1	Scratch	**HIGH**	86%	73,335 - 452,559	YOLO_SEG	PENDING
2	Discolouration	**MEDIUM**	80%	87,53 - 436,310	OPENCV	PENDING
3	Discolouration	**LOW**	69%	562,355 - 641,479	OPENCV	PENDING
4	Discolouration	**LOW**	67%	498,204 - 625,275	OPENCV	PENDING
5	Scratch	**LOW**	66%	459,112 - 486,161	OPENCV	PENDING

ROOT-CAUSE HINTS

Discolouration (3)
• Inconsistent curing, heat treatment, or coating thickness
• Oxidation / corrosion from moisture exposure or contamination
• Raw-material or paint batch variation
→ Check oven / curing temperature uniformity and dwell time
→ Review storage humidity and batch traceability of coating material

Scratch (2)
• Abrasive contact during conveyor transfer or manual handling
• Burrs or debris on guide rails, fixtures, or tooling
• Insufficient protective film / interleaving between stacked parts
→ Inspect and deburr guide rails, grippers, and transfer chutes
→ Add protective film or separators at stacking and packing stations

DETERMINATION & SIGN-OFF

Inspector Remarks: Automated computer-vision inspection; detections are subject to inspector confirmation.

Analysed by: Defect Inspection Engine (YOLO_SEG+ANOMALY+OPENCV)

Quality Sign-off:
_____